

Session

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Math Coaching Assistant

 $4x + 3 = 2x + 11$ 

To solve for x , we should first group the terms with x on one side of the equation. The current equation is $4x + 3 = 2x + 11$. What is the resulting equation if we subtract $2x$ from both sides?

 $2x + 3 = 11$ 

That's correct. The equation is now $2x + 3 = 11$. What is the resulting equation if we subtract 3 from both sides?



But why do we subtract $2x$ from both sides?



We do this to keep the equation balanced while moving all terms with x to one side. By subtracting $2x$ from both sides, we eliminate it from the right side without changing the truth of the equation. Now that we have $2x + 3 = 11$, what is the next step to get x by itself?



I dont understand what you're talking about

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I dont understand what you're talking about



Think of an equation like a balanced scale where the '=' means both sides weigh the same. To keep the equation $2x + 3 = 11$ balanced while we remove the '+ 3' from the left, what must we do to the 11 on the right?



Add 3 to 11



That is incorrect. If we have a scale and we take away 3 pounds from the left side, what must we do to the right side to keep it perfectly level?



Take away 3 pounds from the right side



Exactly. To keep the equation balanced, we perform the same operation on both sides. Starting with $2x + 3 = 11$, what do we get after subtracting 3 from both sides?



what does this do with a scale



The scale represents the equation: the left side is $2x + 3$ and the right side is 11. Subtracting 3 from both sides is like removing 3 pounds from each side of the scale to keep it balanced. Given

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The scale represents the equation: the left side is $2x + 3$ and the right side is 11. Subtracting 3 from both sides is like removing 3 pounds from each side of the scale to keep it balanced. Given the equation $2x + 3 = 11$, what is the result after subtracting 3 from both sides?



$2x = 8$



That's correct. Now the equation is $2x = 8$. To find the value of x , what should we do to both sides of the equation?



Divide by 2



That's correct. The equation is currently $2x = 8$. What is the value of x after dividing both sides by 2?



$x = 4$



That's correct! You've solved the equation. To find $x = 4$, we first grouped the x terms by subtracting $2x$ from both sides to get $2x + 3 = 11$. Then, we balanced the equation by subtracting 3 from both sides to isolate the x term, resulting in $2x = 8$. Finally, we divided by 2 to find the value of x . Remember, an equation is like a scale: any operation performed on one side must be performed on the other to keep it balanced.

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